

“IF A MAN WILL **BEGIN WITH CERTAINTIES**, HE SHALL **END IN DOUBTS**; BUT IF HE WILL BE CONTENT TO **BEGIN WITH DOUBTS**, HE SHALL **END IN CERTAINTIES**. ”

Francis Bacon, English philosopher, from *The Advancement of Learning*, 1605

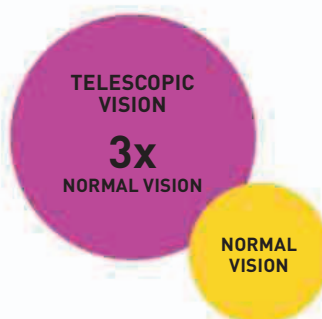
AT THE BEGINNING OF THE 17TH CENTURY, SCIENCE

was still known as “Natural Philosophy.” In 1605, however, English philosopher **Francis Bacon** (1561–1626) published his first work, *The Advancement of Learning*, setting out arguments for using **induction**—a process of drawing conclusions from data accumulated by observation—as the basis for scientific knowledge. Later known as the **Baconian method**, or “scientific method,” induction became important in modern experimental science.

German astronomer **Johannes Kepler** observed the appearance of a comet (now known as **Comet**

Halley) in 1607, and noted its position and path across the night sky. He realized that it was traveling well outside the orbit of the Moon. His observations later influenced his laws of planetary motion (see pp.100–101).

The invention of the first two-lens telescope is generally attributed to Dutch inventor **Hans Lippershey** (c.1570–1619) in 1608. Unlike later reflecting telescopes that used mirrors, Lippershey’s refracting telescope had a lens at each end. Although Lippershey could not obtain a patent for it, his invention earned him money and recognition for its military and commercial uses.



Magnification of early telescopes

The telescopes built by Lippershey and his contemporaries were capable of magnifying an image to about three times its size.

Another invention that was to revolutionize military campaigns was the **flintlock mechanism** for firearms. Probably the work of gunsmith and violin-maker **Marin le Bourgeois** (c.1550–1634), the flintlock appeared around 1608 in France. Quicker and more efficient than previous mechanisms, the flintlock was also safer because it could be locked into position during reloading. It remained in use for more than 200 years.

Lippershey in his workshop

In creating his refracting telescope, Hans Lippershey may have used a combination of concave and convex lenses, or two convex lenses.



Around 1,344 light-years away and situated in the constellation of Orion, the Orion nebula is one of the brightest and closest nebulae to Earth.

IN 1609, GERMAN ASTRONOMER JOHANNES KEPLER’S

book *Astronomia Nova* (*New Astronomy*) was published, describing his observations of the motion of the planet Mars. His detailed measurements and calculations confirmed the theory that the **planets revolved around the Sun**, and went further in suggesting that they did so in **elliptical, rather than circular, orbits**. He also pointed out that the **speed** at which they orbited the Sun did not remain constant but **changed according to their position on the orbit**.

These principles formed the basis of the first two of the three laws that are now known as **Kepler’s laws of planetary motion** (see pp.100–101). The first law states that each planet has an elliptical orbit with the Sun as one of the focuses; the second that the speed of a planet is inversely proportional to its distance from the Sun—that is, a planet moves fastest when it is closest to the Sun.

News of the invention of the refracting telescope reached Italy in 1609, and Galileo Galilei set about building one for himself. **Galileo’s telescope** allowed him to make detailed astronomical observations. His early telescopes had a magnification of about eight times normal vision, but he later improved the design

4 THE NUMBER OF MOONS OF JUPITER OBSERVED BY GALILEO

to around 30 times magnification. English astronomer **Thomas Herriot** (1560–1621) had used a telescope to study the Moon in 1609, and had produced the first drawings of its surface. The following year, Galileo used his superior telescope and artistic training to produce **detailed maps of the lunar landscape**, clearly showing the irregularities to be craters and mountains. So accurate were his maps that he was even able to estimate the height of the mountains on the surface of the Moon.

Galileo was also able to **examine other planets**, and in 1610 turned his attention to **Jupiter**. He noticed three



1605 Francis Bacon publishes *The Advancement of Learning*

1607 Johannes Kepler records the appearance and motion of a comet, later known as Comet Halley

1608 The flintlock mechanism for firearms first appears

1608 Hans Lippershey builds a refracting telescope with two lenses

1609 Galileo Galilei demonstrates one of his early telescopes in Venice

1609 In *Astronomia Nova*, Johannes Kepler states his first two laws of planetary motion

1609 Dutch inventor Cornelis Drebbel invents a thermostat